Checkpoint 4 — Synthetic Data Strategy

Anomaly Detection Platform — Amen Bank

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1 Scope of this Checkpoint

This checkpoint records the design decisions made for the **synthetic data generation** component of the platform. It is a delta on top of Checkpoints 3.1–3.4 (architecture, profiles, enriched events). No prior decision is overridden; this checkpoint extends the data layer with the strategy for producing training and evaluation data.

2 Two Distinct Tools for Two Distinct Purposes

A previously identified conflation is resolved: synthetic data and live event simulation are **two separate tools**.

Aspect	Training Data Generator	Live Event Simulator
Tool	Faker / SDV + custom logic	Custom streaming simulator
Format	Static CSV / Parquet	RedPanda events in real time
Purpose	Model training and evaluation	End-to-end pipeline demo
Anomaly injection	Controlled, labeled	Configurable scenarios
Scope of this checkpoint	**In scope**	Out of scope (later)

3 Calibration Strategy — Option C

3.1 Decision

The training data generator uses a **calibrated synthetic** approach:

- **Generator architecture**: designed in-house (this project).
- **Generator parameters**: extracted from published real-data sources (banking fraud datasets, academic papers, regulatory publications).

3.2 Rationale

Two alternative approaches were rejected:

- **Pure synthetic** (parameters assumed): risk of synthetic-to-real distribution shift; results not defensible.
- **Real dataset adapted**: schema mismatch with the dual-actor teller-operations context; no public dataset covers this domain; introduces a circular validation trap.

The chosen approach gives us full control over the schema and anomaly labels while grounding every parameter in published evidence. The governing principle:

> *A system that detects anomalies in unrealistic data is a system that detects nothing real. Calibration is not optional — it is what makes the work valid.*

4 Three-Layer Generator Architecture

The generator produces events through three nested layers that create realistic intra-client coherence and inter-client diversity.

4.1 Layer 1 — Archetype Prior

For each of the five archetypes (salaried, small biz, student, retiree, big biz), define the population-level distributions for:

- Operation type mix (proportion across the 4 cash operations)
- Transaction frequency (count per time unit)
- Amount distribution **per operation type**
- Temporal patterns (hour of day, day of week, payday effects)
- Spatial patterns (branch concentration)
- Relational patterns (beneficiary / emitter known-vs-new rate)
- Channel mix (guichet vs GAB)
- Inter-event time distribution

4.2 Layer 2 — Per-Client Personality

When a client is instantiated, their specific parameters are sampled *once* from the archetype prior:

- Preferred branch (primary + secondary)
- Salary day (or pension day, or business cycle day)
- Typical amount distribution centered on this client's mean
- Recurring beneficiaries / emitters (small fixed set)

This layer creates the **intra-client coherence** that the LSTM needs to learn meaningful sequential patterns. Without it, every client collapses into a noisy copy of the archetype mean.

4.3 Layer 3 — Per-Event Noise

Each event is sampled from the client's personality with small controlled noise:

- Amount sampled from the client's distribution
- Branch is primary ∼85% of the time, secondary otherwise
- Beneficiary drawn from the recurring set ∼90% of the time, new one otherwise

5 Implications for LSTM Training

The LSTM is reframed as a **next-step prediction model**, not a classifier. Given the previous $N - 1$ events of a client, it predicts properties of the N-th event. Anomaly score = prediction error.

5.1 Multi-Task Prediction Head

Target	Type	Notes
`operation_type`	Softmax (4 classes)	Core target
`amount_bucket`	Softmax (∼6 buckets)	Includes 8–10k DT smurfing band as its own bucket
`channel`	Softmax (guichet/GAB)	Low loss weight
`branch_known`	Sigmoid binary	Known-vs-new for this client
`employee_known`	Sigmoid binary	Known-vs-new
`beneficiary_known`	Sigmoid binary	Virements only
`emitter_known`	Sigmoid binary	Remise chèques only

5.2 Sequence-Length Constraint

Training sequence length must be $\leq$ the inference buffer size. The client profile's `recent_events_buffer` is capped at 50 events; therefore training sequences are also length 50. Training on longer sequences than the buffer would teach the model to rely on long-range dependencies that do not exist at inference time.

6 Separation of Concerns: LSTM vs Decision Service

The LSTM (in the Scoring Service) produces a **continuous, statistical anomaly score**. It contains no regulatory logic.

LAB/FT regulatory thresholds (e.g. the 10 000 DT reporting threshold) live in the **Decision Service** as deterministic rules. The alert tier (INFO / REVIEW / ALERT / BLOCK) is a function of both inputs:

Statistical score	Regulatory flag	Alert tier
Low	None	INFO
High	None	REVIEW
Low	Triggered	REVIEW
High	Triggered	ALERT
Very high	Triggered, large amount	BLOCK

This separation makes the ML model portable (different jurisdictions need different rules, not different models) and the rules auditable (regulators can be answered with deterministic logic, not neural networks).

7 Source Extraction Template v1

To make Option C operational, every calibration source is processed through a standardised extraction template. Filling this template once per source simultaneously addresses (a) parameter extraction, (b) anomaly base rates, and (c) archetype mapping.

7.1 Trust Metadata (7 dimensions)

- **Recency** — publication / collection year
- **Authority** — who produced it (central bank, peer-reviewed lab, individual)
- **Peer review** — peer-reviewed paper / preprint / dataset / blog
- **Sample size** — number of transactions or clients
- **Geographic relevance** — MENA region preferred, then Europe, then elsewhere
- **Data type** — real anonymized / simulated calibrated / pure synthetic / theoretical
- **Reproducibility** — data accessibility + methodology documentation

7.2 Coverage Matrix

For each source, fill the following matrix with confidence per cell (High / Medium / Low / Not covered):

Dimension	Per op type?	Value	Confidence
Amount distribution	yes	e.g. LogNormal(μ,σ)	—
Operation type mix	no	e.g. 60/20/15/5	—
Transaction frequency	yes	per week / month	—
Inter-event time	yes	gap distribution	—
Temporal patterns	yes	hour / weekday	—
Spatial / branch patterns	no	concentration	—
Channel mix	no	guichet / GAB	—
Beneficiary / emitter	partial	known-rate	—

7.3 Anomaly Typology Breakdown

Per source, document anomaly base rates by typology (smurfing, doublons, montant hors norme, account takeover, etc.) along with the source's definition / labelling criteria.

7.4 Archetype Mapping and Provenance

Per source, record which archetype(s) it informs, the strength of match, and a paragraph-length provenance note for the final report.

8 Parameter Ledger (Cross-Source Synthesis)

To handle disagreements between sources and to track the path from extracted data to generator parameters, a project-level **parameter ledger** is maintained. Each row of the ledger corresponds to one generator parameter and contains:

- Parameter name and target archetype
- All sources informing it, with their proposed values
- Chosen value and resolution rationale
- Conflict log (if sources disagree)
- Sensitivity analysis range (for robustness testing)

Conflict resolution strategy combines: **trust hierarchy** (higher-tier source wins), **domain-relevance weighting** (MENA proximity wins), and **range-based sensitivity testing** (when disagreement persists, both bounds are tested during model evaluation).

9 Next Steps

1. Extract **PaySim** as the first calibration source using template v1.
2. Decide the shortlist of remaining sources to extract (Banksim, BCT publications, GAFI typologies, others).
3. Initialise the parameter ledger and populate it as sources are extracted.
4. Once ≥ 4 sources are extracted, encode Layer 1 (archetype priors) and begin generator implementation.

10 Open Items

- Final injection rate for anomalies (realistic ∼0.1–1% vs. training-friendly ∼1–5%) — to be decided after source extraction.
- Total volume of generated data (client count, time span) — deferred until Layer 1 is calibrated.
- Score fusion across the seven LSTM prediction heads — deferred to scoring service design.